

HEALTHCONNECT CLINIC

FINAL ANALYTICS & DECISION SUPPORT PACKAGE

Week 8 • Data Analytics Track

Improving Patient Appointment Attendance and Healthcare Support Using Data & AI

Project stage	Final Integration → Project Readiness → Presentation
Primary contribution	Final analytical evidence and decision support
Evidence window	Weeks 5–7 validated project work
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1. Executive Summary

Week 8 consolidates the HealthConnect Data Analytics work developed from Weeks 5–7 into a final decision-support contribution. The analysis identifies missed appointments as a substantial operational issue and highlights three practical areas for attention: previous no-show history, follow-up appointments, and reminder coverage. The final interpretation also preserves the methodological lesson from cross-track validation: KPI definitions and treatment of Cancelled appointments materially affect measured rates.

The final analytical position is deliberately evidence-based. The findings describe associations rather than causal effects, and the Week 7 DAX-dependent validation tests that became unstable in Power BI are not presented as completed tests. The result is a decision-support package that is useful for segmentation, prioritisation and future modelling while remaining transparent about what has and has not been validated.

2. Week 7 → Week 8 Transition

Transition item	Final Week 8 position
Main Week 7 output	Validation and refinement of the Week 6 decision-support analysis.
Most important testing result	The established KPI and segmentation findings remained the working evidence base; reminder differences were interpreted cautiously across outcome definitions.
Major issue discovered	Power BI DAX instability prevented completion of outcome reconciliation, percentage reconciliation, Follow-up filter testing and dashboard-to-source verification.
What was refined	KPI definitions and the interpretation of reminder effects were kept explicit; unsupported test results were not claimed.
Component ready	The analytical findings, recommendations and cross-track evidence are ready to support final decision-making and presentation.
Still to integrate	The final analytics contribution must be connected to Data Science interpretation and the wider HC-POD solution.
Tracks involved	Data Analytics ↔ Data Science, with findings also supporting Project Management and the wider final integration.
Final demonstration	Show how the data identified priority segments, how cross-track validation refined interpretation, and how the evidence supports practical HealthConnect decisions.

3. Final Analytics Integration Readiness

Requirement	Response
Final component	HealthConnect Final Analytics & Decision Support Package: KPIs, validated findings, visual evidence, business insights and recommendations.
Problem addressed	Reducing missed appointments and improving appointment attendance and use of available appointment capacity.

Current status	Ready as a decision-support and presentation component, with clearly documented validation limitations.
Week 7 improvement	Refined the interpretation of reminder effects and preserved explicit KPI definitions; documented pending DAX-dependent tests.
Dependencies	Data Science for modelling interpretation; shared definitions across tracks; Project Management for final integration/presentation.
Outputs provided	Final KPIs, analytical findings, segmentation insights, recommendations and limitations.
Evidence of readiness	Consistent Week 5–7 evidence, Power BI analytical outputs, and documented cross-track methodological refinement.
Remaining limitations	Observational evidence, missing values, cancelled-appointment treatment differences, small extreme-history groups, and unresolved DAX-dependent validation tests.

4. Final KPI Set

KPI	Final result	Interpretation
Attendance rate	46.28%	Share of all appointments marked Attended.
Overall no-show rate	48.46%	Share of all appointments marked No-show.
Cancelled appointments	5.26%	Share of all appointments marked Cancelled.
Attended vs No-show no-show rate	≈51.15%	No-show share when the comparison is restricted to Attended and No-show outcomes.
Follow-up no-show rate	51.23%	Highest observed no-show rate among the displayed appointment types.

Important definition note: the 48.46% overall no-show rate and the approximately 51.15% Attended-vs-No-show rate are different KPIs with different denominators. They should not be presented as contradictory figures.

5. Key Analytical Findings

01 — Missed appointments are the core operational issue

No-shows account for 48.46% of all appointments, exceeding the 46.28% attendance rate.

02 — Previous no-show history is the strongest segmentation signal

The displayed analysis shows an upward no-show pattern as previous no-shows increase, reaching approximately 97% for the group with five previous no-shows. This extreme group requires sample-size verification before being treated as a stable population estimate.

03 — Follow-up appointments need targeted attention

Follow-ups have the highest displayed appointment-type no-show rate at 51.23%, above the 48.46% overall no-show rate.

04 — Reminder status is relevant, but channel performance is not universal

Overall no-show rates were 51.39% for No Reminder, 48.41% for Email, 45.75% for SMS and 49.77% for WhatsApp. However, performance varies by appointment type, so no single channel should be declared universally superior.

05 — Outcome definitions materially affect interpretation

Data Science excluded 263 Cancelled appointments when comparing Attended versus No-show outcomes. This produced a smaller reminder-related difference of approximately five percentage points and reinforced the need for common KPI definitions.

06 — Distance and waiting time remain secondary

The Week 5 analysis did not identify meaningful relationships for these variables, so they are not elevated to primary drivers in the final analytics position.

6. Reminder Analysis Snapshot

Reminder category	No-show rate
No Reminder	51.39%
Email	48.41%
SMS	45.75%
WhatsApp	49.77%

These figures support segmentation and investigation, not a causal claim that one reminder channel causes better attendance. Channel performance differs by appointment type.

7. Business Insights & Recommendations

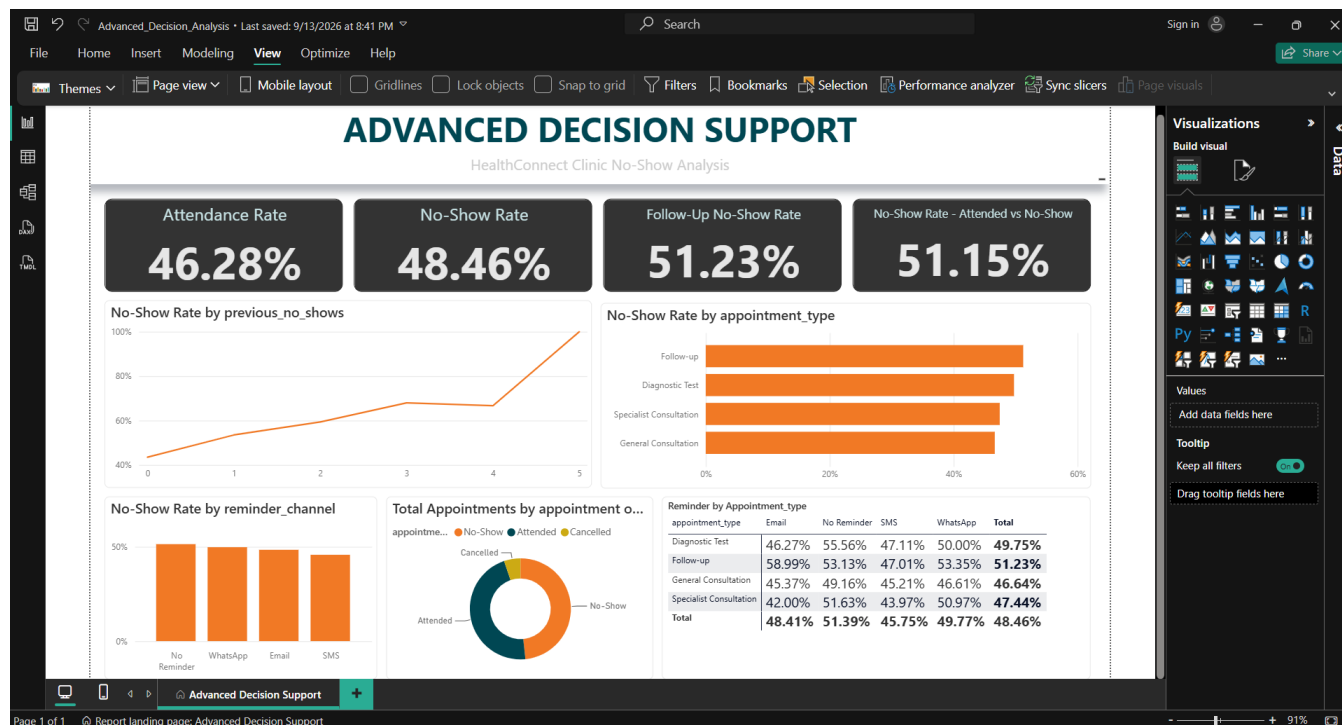
Recommendation	Decision value
Prioritise repeat no-show patients	Use previous no-show history as a practical risk-segmentation variable for targeted attendance-support interventions.
Improve reminder coverage	Identify appointments with no recorded reminder and improve reminder-data completeness and coverage.
Segment reminder strategy	Compare reminder channels within appointment types rather than assuming one channel is best for everyone.
Investigate follow-up appointments	Review scheduling, communication, patient context and operational factors associated with the elevated follow-up no-show rate.
Standardise KPI definitions	Use a shared population definition, outcome taxonomy, missing-value treatment and denominator rules across Analytics and Data Science.
Keep weak variables secondary	Do not treat distance or waiting time as primary explanations unless future evidence shows a stronger relationship.

8. Cross-Track Contribution

Data Analytics worked most directly with Data Science. The collaboration identified a methodological difference: Analytics retained Cancelled appointments in the primary dataset, while Data Science excluded 263 Cancelled appointments before comparing Attended versus No-show outcomes. The difference changed the denominator and reduced the apparent reminder effect to approximately five percentage points.

The practical result was not to force one analysis to match the other, but to make the definitions explicit and provide a comparable Attended-vs-No-show KPI. The final contribution to Data Science is therefore a set of validated analytical signals—previous no-show history, reminder information and appointment type—together with clearer interpretation boundaries.

9. Final Dashboard / Visualization Guidance



The final dashboard/report should foreground the following visuals:

Four KPI cards: Attendance 46.28% | Overall No-show 48.46% | Cancelled 5.26% | Follow-up No-show 51.23%.

Bar chart: No-show rate by appointment type.

Line chart: No-show rate by previous no-shows, with a clear caution that the five-history group requires sample-size verification.

Reminder-channel comparison: No Reminder, Email, SMS and WhatsApp.

Segmented reminder matrix by appointment type, showing why channel performance should not be generalised.

Presentation principle: prioritise clarity over the number of visuals. Each chart should answer a decision question and should not imply causation where the analysis only demonstrates association.

10. Analytical Limitations & Risk Controls

Limitation	Control / implication
Observational evidence	Associations do not establish that reminders, appointment type or previous no-shows cause missed appointments.
Outcome-definition sensitivity	Different treatment of Cancelled appointments changes denominators and measured rates.
Missing values	Reminder-channel completeness differs across processing approaches and can affect interpretation.
Extreme history group	The approximately 97% rate for five previous no-shows may reflect a relatively small group and needs further validation.
Power BI validation status	Week 7 DAX-dependent outcome reconciliation, percentage reconciliation, Follow-up filter testing and dashboard-to-source verification remained pending because the table became unstable.
Model readiness	This package is decision support, not a production predictive model.

11. Final HealthConnect Value

The Analytics contribution helps HealthConnect move from a broad attendance problem to a focused set of decision areas. Rather than treating every patient or appointment equally, the evidence supports attention to repeated no-show history, follow-up appointments and reminder coverage. The work also improves the reliability of the wider solution by making KPI definitions and analytical limitations visible to Data Science and the other project tracks.

In the final integrated solution, Analytics provides the evidence layer: what is happening, where the strongest observed patterns are, how reliable the interpretation is, and which areas deserve further modelling or operational investigation.

12. Final Executive Takeaway

HealthConnect's attendance problem is substantial: nearly half of all appointments are recorded as no-shows. The strongest actionable signals are previous no-show history, follow-up appointment type and reminder coverage. However, the evidence is observational, and the treatment of Cancelled appointments materially changes some measured effects. The final recommendation is therefore targeted, segmented and evidence-led not causal or universal.